

THE ORAL DEFENSE & RELEASE VERDICT QUALIFICATION GATE

Auditing the 11-artifact Cumulative Design Dossier across 4 review personas for a binding release verdict

1. EMBEDDED & SILICON LEAD • Zero-copy DMABUF ring buffers. • PTP Exposure midpoint skew < 5 μs. • FreeRTOS 1 kHz deterministic tick. • Watchdog countdown < 50 ms. • Zero dynamic memory allocation. STATUS: SIGNED ✓ P99 = 24.2 ms Bounded	2. EMBODIED ML SPECIALIST • Metric SE(3) spatial affordances. • Expiring intent lease TTL (500 ms). • C^2 quintic polynomial spline blend. • Governed episode failure slicing. • Bounded KV-cache memory. STATUS: SIGNED ✓ Zero Causal Collapse	3. SYSTEM SAFETY LEAD • CBF forward invariance $h(x) \geq 0$. • Dynamic stopping $d_{stop} \leq d_{clear}$. • IEC 60204-1 Cat 0/1/2 fallback. • 5,000 seeded fault injection runs. • Complete STPA hazard matrix. STATUS: SIGNED ✓ 0 Physical Breaches	4. PRACTITIONER UX LEAD • Clean, unambiguous C contracts. • Reproducible bench realization. • Manageable 6 ECTS workload. • Zero undocumented magic numbers. • Complete diagnostic logs. STATUS: SIGNED ✓ 100% Bench Reproducible
★ VERDICT 1: DEPLOY Full ODD Release Authorized • Passes all 4 qualification rungs. • 100% fault containment verified. • Unanimous 4-expert approval. RESULT: Production Shipment	⚠ VERDICT 2: CONDITION Restricted ODD Operation • Speed derated to 0.5 m/s. • Human co-pilot required ($\alpha \leq 0.5$). • Requires secondary audit within 30d. RESULT: Supervised Pilot	✖ VERDICT 3: REFUSE Shipment Blocked • Tail latency exceeds τ_{world}. • Fault injection boundary breach. • Missing empirical bench proofs. RESULT: Redesign Required	